Eigenvalue Positivity for the Hecke Staircase Element via Kazhdan–Lusztig Positivity

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Abstract

We prove that all nonzero eigenvalues of the Hecke staircase element $\Pi_q = \prod_{j=1}^{m}(1 + T_{i_j})$ acting on any Specht module V_λ are positive for all $q > 0$. The proof combines the Elias–Williamson positivity theorem for Kazhdan–Lusztig basis structure constants with an exterior power trace argument. As a corollary, the rank of Π_q on V_λ is constant for $q > 0$. This note is intended to be self-contained: we include the necessary definitions of the Iwahori–Hecke algebra, Specht modules, W-graphs, and the staircase element.

1 The Iwahori–Hecke algebra

Let $n \geq 2$ and let q be a formal parameter (or a positive real number).

Definition 1 (Iwahori–Hecke algebra)**.** The *Iwahori–Hecke algebra* $H_q(S_n)$ is the associative $\mathbb{Z}[q, q^{-1}]$-algebra with generators $T_1, \ldots, T_{n-1}$ subject to the relations:

(H1) *Quadratic relation:* $T_i^2 = (q-1)\,T_i + q \qquad$ for $1 \leq i \leq n-1$.

(H2) *Braid relations:* $T_i\,T_{i+1}\,T_i = T_{i+1}\,T_i\,T_{i+1} \qquad$ for $1 \leq i \leq n-2$.

(H3) *Far commutativity:* $T_i\,T_j = T_j\,T_i \qquad$ for $|i-j| \geq 2$.

At $q = 1$, the quadratic relation becomes $T_i^2 = 1$, so $H_1(S_n) \cong \mathbb{Z}[S_n]$ via $T_i \mapsto s_i$ (the simple transposition $(i,\, i{+}1)$). For general q, $H_q(S_n)$ is a free $\mathbb{Z}[q, q^{-1}]$-module of rank $n!$, with basis $\{T_w : w \in S_n\}$ indexed by $T_w = T_{i_1} \cdots T_{i_\ell}$ for any reduced expression $w = s_{i_1} \cdots s_{i_\ell}$.

Remark 2. We will also use the parameter v with $v^2 = q$. The *Kazhdan–Lusztig generator* is $C'_s = v^{-1}(1 + T_s)$ for a simple reflection s. This normalization is chosen so that the KL basis elements $\{C'_w : w \in S_n\}$ have positivity properties.

2 Specht modules

Definition 3 (Specht modules)**.** A *partition* $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell > 0)$ of n (written $\lambda \vdash n$) indexes a *Young diagram*: an array of n boxes with λ_i boxes in row i. A *standard Young tableau* (SYT) of shape λ is a filling of the Young diagram with $1, \ldots, n$, each appearing once, such that entries increase along each row and down each column.

The *Specht module* V_λ is an irreducible representation of $H_q(S_n)$ (over $\mathbb{Q}(q)$, or over $\mathbb{Q}$ when $q = 1$ gives the classical Specht module for S_n). Its dimension is $f^\lambda = |\{\text{SYT of shape } \lambda\}|$, computed by the hook-length formula. The Specht modules $\{V_\lambda : \lambda \vdash n\}$ form a complete set of pairwise non-isomorphic irreducible $H_q(S_n)$-modules (for generic q, i.e. q not a root of unity).

At $q = 1$, these are the classical irreducible representations of S_n. For general q, Specht modules arise as *cell modules* of the Kazhdan–Lusztig cell structure: each two-sided cell corresponds to a partition, and the cell module is the Specht module V_λ.

3 W-graphs and positivity

The key input to our proof is the non-negativity of certain matrix entries in the Kazhdan–Lusztig basis. This is most naturally expressed through the theory of W-graphs.

Definition 4 (W-graph [KL79])**.** A *W-graph* for $H_q(S_n)$ is a triple (V, I, μ) where:

- V is a finite set of vertices (forming a basis of the representation),
- $I\colon V \to 2^{\{1,\dots,n-1\}}$ assigns a *descent set* $I(x) \subseteq \{1, \dots, n-1\}$ to each vertex,
- $\mu\colon V \times V \to \mathbb{Z}[v^{-1}]$ assigns edge weights,

such that the action of T_i on the basis $\{e_x : x \in V\}$ is:

$$T_i \cdot e_x \;=\; \begin{cases} q \cdot e_x & \text{if } i \notin I(x), \\ -e_x + \displaystyle\sum_{\substack{y \in V \\ i \notin I(y)}} \mu(y,x)\, v \cdot e_y & \text{if } i \in I(x). \end{cases}$$

Theorem 5 (Kazhdan–Lusztig [KL79], Elias–Williamson [EW14])**.** *Every Specht module V_λ admits a W-graph in which all edge weights satisfy $\mu(y,x) \in \mathbb{Z}_{\geq 0}[v^{-1}]$. More generally, the structure constants for the product of Kazhdan–Lusztig basis elements satisfy*

$$C'_u \cdot C'_v \;=\; \sum_w h^w_{u,v}(v)\, C'_w, \qquad h^w_{u,v} \in \mathbb{Z}_{\geq 0}[v, v^{-1}].$$

The non-negativity of μ-coefficients was conjectured by Kazhdan–Lusztig [KL79] and proved by Elias–Williamson [EW14].

The crucial consequence for us is:

Proposition 6. *In the W-graph basis of V_λ, the matrix of $1 + T_i$ has all entries in $\mathbb{Z}_{\geq 0}[v]$. Specifically:*

$$(1 + T_i) \cdot e_x \;=\; \begin{cases} (1+q)\, e_x & \textit{if } i \notin I(x), \\ \displaystyle\sum_{\substack{y \in V \\ i \notin I(y)}} \mu(y,x)\, v \cdot e_y & \textit{if } i \in I(x). \end{cases}$$

Since $\mu(y,x) \in \mathbb{Z}_{\geq 0}[v^{-1}]$, the product $\mu(y,x) \cdot v \in \mathbb{Z}_{\geq 0}[v^{-1}] \cdot v \subseteq \mathbb{Z}_{\geq 0}[v, v^{-1}]$. In fact, $1 + T_i = v \cdot C'_{s_i}$ and the entries lie in $\mathbb{Z}_{\geq 0}[v]$ (no negative powers of v survive because C'_{s_i} acts on the cell module with Laurent polynomial entries in $\mathbb{Z}_{\geq 0}[v^{-1}]$ and the extra factor of v shifts all exponents up by 1).

4 The staircase element

The long element $w_0 \in S_n$ is the unique permutation of maximal length $m = \binom{n}{2}$. A *reduced word* for w_0 is a sequence $(i_1, \ldots, i_m)$ with $w_0 = s_{i_1} \cdots s_{i_m}$.

Definition 7 (Staircase reduced word)**.** The *staircase reduced word* for w_0 is:

$$\underbrace{s_1}_{k=1},\ \underbrace{s_2,\ s_1}_{k=2},\ \underbrace{s_3,\ s_2,\ s_1}_{k=3},\ \ldots,\ \underbrace{s_{n-1},\ s_{n-2},\ \ldots,\ s_1}_{k=n-1}.$$

The k-th block is $(s_k,\ s_{k-1},\ \ldots,\ s_1)$, which inserts element $k+1$ into positions 1 through $k+1$. This is a reduced word for w_0 because it sorts $(1, 2, \ldots, n)$ to $(n, n-1, \ldots, 1)$ by successively moving each new largest element to the front.

Definition 8 (Hecke staircase element)**.** The *Hecke staircase element* is:

$$\Pi_q \;=\; \prod_{k=1}^{n-1} \prod_{j=k}^{1} (1 + T_j) \;=\; \prod_{j=1}^{m} (1 + T_{i_j}) \;=\; v^m \prod_{j=1}^{m} C'_{s_{i_j}},$$

where $(i_1, \ldots, i_m)$ is the staircase reduced word.

Remark 9 (Why this reduced word?)*.* The staircase word has several special properties:

1. Its reversal is also a reduced word for w_0 (the "anti-staircase"), which implies $f_w = f_{w^{-1}}$ in the $q = 1$ group algebra expansion.

2. The block structure $(1+T_k)(1+T_{k-1}) \cdots (1+T_1)$ at step k is the divided symmetrization operator for the parabolic subgroup S_{k+1}. This gives Π_q a cascade structure: it is a composition of partial symmetrizers of increasing rank.

3. At $q = 0$, the factors $(1 + T_i)$ become sorting operators $(1 + s_i)$, and Π_0 implements the well-known "staircase sorting" of permutations.

4. At $q = 1$, $\Pi_1 = \prod(1 + s_{i_j}) \in \mathbb{Q}[S_n]$, and its image on each Specht module V_λ is the H-invariant subspace V_λ^H for the parabolic subgroup $H = \langle s_1, s_3, s_5, \ldots \rangle$.

5 Main results

Theorem 10 (Eigenvalue Positivity)**.** *For every $\lambda \vdash n$ and $q > 0$, all nonzero eigenvalues of Π_q on the Specht module V_λ are strictly positive.*

Corollary 11 (Rank Constancy)**.** $\operatorname{rank}(\Pi_q|_{V_\lambda})$ *is constant for all $q > 0$.*

6 Proof of trace non-negativity

Theorem 12 (Trace Non-negativity)**.** *For every partition $\mu \vdash n$, the trace $\operatorname{tr}_\mu(\Pi_q)$ of Π_q on the Specht module V_μ is a polynomial in q with non-negative integer coefficients.*

Proof. Step 1 (Product expansion via Elias–Williamson). By Theorem 5, the product of Kazhdan–Lusztig basis elements has non-negative expansion:

$$\prod_{j=1}^{m} C'_{s_{i_j}} \;=\; \sum_{w\in S_n} a_w(v)\, C'_w \qquad \text{with } a_w \in \mathbb{Z}_{\geq 0}[v].$$

This follows because the product $C'_{s_{i_1}} \cdot C'_{s_{i_2}} \cdots C'_{s_{i_m}}$ is built up one factor at a time, and at each step, multiplying by $C'_{s_{i_j}}$ preserves non-negativity of the expansion coefficients by the Elias–Williamson positivity of the structure constants $h^w_{u,v}$.

Step 2 (Trace of a KL basis element on a cell module). Let V_μ be the cell module (Specht module) corresponding to μ. In the KL cellular basis $\{\bar{C}_x : x \in \mathrm{cell}(\mu)\}$, left multiplication by C'_w gives:

$$C'_w \cdot \bar{C}_x \;=\; \sum_{y\in \mathrm{cell}(\mu)} h_{w,x,y}(v)\, \bar{C}_y$$

where $h_{w,x,y} \in \mathbb{Z}_{\geq 0}[v, v^{-1}]$ by KL positivity. (For cell modules, in fact $h_{w,x,y} \in \mathbb{Z}_{\geq 0}[v^{-1}]$.) The trace is therefore:

$$\mathrm{tr}_\mu(C'_w) \;=\; \sum_{x\in \mathrm{cell}(\mu)} h_{w,x,x}(v) \;\in\; \mathbb{Z}_{\geq 0}[v, v^{-1}].$$

Step 3 (Combining and specializing). Since $\Pi_q = v^m \prod_j C'_{s_{i_j}}$:

$$\mathrm{tr}_\mu(\Pi_q) \;=\; v^m \sum_w a_w(v) \cdot \mathrm{tr}_\mu(C'_w) \;\in\; \mathbb{Z}_{\geq 0}[v, v^{-1}].$$

Now, $\mathrm{tr}_\mu(\Pi_q)$ lies in $\mathbb{Z}[q] = \mathbb{Z}[v^2]$ because it is the trace of a $\mathbb{Z}[q]$-matrix on a $\mathbb{Z}[q]$-module (the representation matrices in the T-basis have entries in $\mathbb{Z}[q]$). So all odd powers of v vanish. A polynomial in $\mathbb{Z}_{\geq 0}[v, v^{-1}] \cap \mathbb{Z}[v^2]$ has non-negative coefficients when re-expressed in $q = v^2$. Hence $\mathrm{tr}_\mu(\Pi_q) \in \mathbb{Z}_{\geq 0}[q]$. □

7 Proof of eigenvalue positivity

Proof of Theorem 10. Step 1 (Elementary symmetric functions of eigenvalues). Let r_λ denote the rank of Π_q on V_λ over $\mathbb{Q}(q)$, and let $\alpha_1(q), \ldots, \alpha_{r_\lambda}(q)$ be the nonzero eigenvalues (as algebraic functions of q). For $1 \leq k \leq r_\lambda$, let

$$e_k(q) \;=\; \sum_{1\leq j_1<\cdots<j_k\leq r_\lambda} \alpha_{j_1}(q) \cdots \alpha_{j_k}(q)$$

be the k-th elementary symmetric function of the nonzero eigenvalues. By the standard identity relating exterior powers to elementary symmetric functions:

$$e_k(q) \;=\; \mathrm{tr}\Big(\textstyle\bigwedge^k (\Pi_q|_{V_\lambda}) \text{ on } \bigwedge^k V_\lambda\Big).$$

This uses the multiplicativity of exterior powers: $\bigwedge^k(AB) = (\bigwedge^k A)(\bigwedge^k B)$, so the product $\Pi_q = \prod(1 + T_{i_j})$ lifts to $\bigwedge^k(\Pi_q) = \prod \bigwedge^k(1 + T_{i_j})$.

Step 2 (Decomposition into Specht modules). As an $H_q(S_n)$-module,

$$\bigwedge^k V_\lambda \;\cong\; \bigoplus_\mu V_\mu^{\oplus a_{k,\mu}}$$

with $a_{k,\mu} \geq 0$. By Theorem 12:

$$e_k(q) \;=\; \sum_\mu a_{k,\mu} \cdot \mathrm{tr}_\mu(\Pi_q) \;\in\; \mathbb{Z}_{\geq 0}[q].$$

Step 3 (Non-vanishing). Over $\mathbb{Q}(q)$, the matrix $\Pi_q|_{V_\lambda}$ has rank r_λ. Hence $e_{r_\lambda} = \alpha_1 \cdots \alpha_{r_\lambda} \neq 0$ as a rational function, and therefore as a polynomial in q (since $e_k \in \mathbb{Z}[q]$). More generally, $e_k \neq 0$ for $1 \leq k \leq r_\lambda$. A nonzero polynomial in $\mathbb{Z}_{\geq 0}[q]$ is strictly positive for all $q > 0$.

Step 4 (Positivity of all eigenvalues). The nonzero eigenvalues $\alpha_1, \ldots, \alpha_{r_\lambda}$ of Π_q on V_λ are real for $q > 0$ (as eigenvalues of a real matrix when q is specialized to a positive real number). They are the roots of the reduced characteristic polynomial:

$$p(x) \;=\; \sum_{k=0}^{r_\lambda} (-1)^k \, e_k(q) \, x^{r_\lambda - k}, \qquad e_0 = 1.$$

We claim $p(x) \neq 0$ for all $x \leq 0$ (when $q > 0$). For $x \leq 0$, consider consecutive terms:

- $(-1)^k \, e_k(q) \, x^{r_\lambda - k}$ has sign $(-1)^k \cdot (+) \cdot (-1)^{r_\lambda - k} = (-1)^{r_\lambda}$.
- $(-1)^{k+1} \, e_{k+1}(q) \, x^{r_\lambda - k - 1}$ has sign $(-1)^{k+1} \cdot (+) \cdot (-1)^{r_\lambda - k - 1} = (-1)^{r_\lambda}$.

All terms have the common sign $(-1)^{r_\lambda}$, so $p(x) \neq 0$ for $x \leq 0$. Therefore all roots of p are positive. □

Proof of Corollary 11. Since all r_λ eigenvalues are strictly positive for $q > 0$, none can vanish, so the rank cannot drop below r_λ. And rank cannot exceed r_λ (the generic rank over $\mathbb{Q}(q)$). □

8 Computational verification

The trace non-negativity and eigenvalue positivity have been verified independently by explicit computation of eigenvalue polynomials for all $\lambda \vdash n$ with $n \leq 8$.

Method. For each $\lambda \vdash n$, we construct the Specht module V_λ using seminormal matrix units, compute the matrix of each factor $(1 + T_{i_j})$, multiply them to obtain $\Pi_q|_{V_\lambda}$, and compute its characteristic polynomial in $\mathbb{Z}[q][x]$. The nonzero eigenvalue polynomials are then extracted.

Sample results.

λ	$\dim V_\lambda$	rank of Π_q	eigenvalue polynomials
(3)	1	1	$(1+q)^3$
$(2,1)$	2	1	$q(1+q)$
$(1,1,1)$	1	1	q^3
(4)	1	1	$(1+q)^6$
$(3,1)$	3	2	$q^2(1+q)^2(1+q+q^2),\ q(1+q)^4$
$(2,2)$	2	1	$q^2(1+q)^2(2q+1)$
$(2,1,1)$	3	2	$q^3(1+q)^2(1+q+q^2),\ q^4(1+q)^2$
(1^4)	1	1	q^6

All listed eigenvalue polynomials have non-negative coefficients and are strictly positive for $q > 0$, confirming Theorems 12 and 10 in these cases. Verification extends through $n = 8$ ($|\{\lambda \vdash 8\}| = 22$ partitions, all confirmed).

Independence from reduced word. Trace non-negativity holds for the staircase word specifically. Other reduced words for w_0 can yield elements Π'_q whose traces have negative coefficients. The staircase word is distinguished by its cascade (parabolic) structure.

9 Remarks

Remark 13 (Minimal hypotheses). The proof requires *only*: (i) the identification $1 + T_s = v \cdot C'_s$, (ii) Elias–Williamson positivity of KL structure constants, and (iii) the decomposition of exterior powers into Specht modules. No case analysis, no explicit eigenvalue computation, no restriction on n, and no assumption on the partition λ.

Remark 14 (Connection to the $q = 0$ limit). At $q = 0$, the Hecke algebra degenerates to the 0-Hecke algebra $H_0(S_n)$, where $T_i^2 = -T_i$. The factors $(1 + T_i)$ become idempotent: $(1 + T_i)^2 = (1 + T_i)$. The staircase element Π_0 acts as the "staircase sorting operator," a projection whose image on V_λ is the H-invariant subspace V_λ^H. Theorem 10 shows that the positivity persists for all $q > 0$, interpolating smoothly from this $q = 0$ projection.

Remark 15 (Application). The eigenvalue positivity theorem is the key input to the "Frobenius squeeze" proof of the H-invariant theorem $\mathrm{rank}(\Pi_q|_{V_\lambda}) = \dim(V_\lambda^H)$: it provides the per-irreducible lower bound that, combined with the total rank upper bound from Frobenius reciprocity, forces equality in every component.

References

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